batmonitor 0.74: charge path (every 5 s) | blue = decision (c/bm_tables.h, enumerated by make check) | green = matrix / amounts | [CBMC] = proved for all inputs

D4 peak window as a forecast export cap (0.49-c: pure function bm_d4)
goal: keep the stacks free for the afternoon peaks, charge only the export above EXPORT_CAP (summer 2026: DO4 on 55 days, 11:45-16:15)

clear-sky model dc(t) (bm_logic.c calc_arrays_kt, from fox2db_logic.h / clearsky.tex)

inputs	
time	unix clock (NTP, Fritzbox)
site	LAT 47.6811 N, LON 11.5732 E (Lenggries)
outdoor temperature	aussen/temp (MQTT, < 15 min old)
steps	
sun	NOAA: elevation, azimuth from time + site
atmosphere	air mass AM = 1 / sin(elev) (≤ 37), Meinel T = 0.7^(AM^0.678)
4 PV strings	tilt / azimuth from south / power, fitted per string (gen_pv_strings.py, 14 clear days): Sofar PV1 west 60 deg / +33 deg / 19430 W, PV2 south 68 deg / -12 deg / 7690 W FoxESS pv2 east 59 deg / -29 deg / 22036 W, pv1 west 67 deg / +32 deg / 2781 W
horizon	trees east: below 23 deg (south-east 15 deg) only 25 % diffuse
month	KT_MONTH empirical: Dec 0.13 ... May 0.91
temperature	cell = outdoor + NOCT 45 x sun; -0.30 %/K above 25 deg C (0.85..1.0)
dc(t) = Σ power x T x kt x cos(incidence) x shade x temperature factor	

dc(t)

inputs
PCC (+ = export), PCC 5 min mean, Sofar Bat1 W (+ = charge), Multis AC-in,
outdoor temperature, BMS SoC / U / CCL, time (local, solar noon)

PCC

export above 18 kW: who takes it (user 2026-10-10: boost and air conditioning before DO4)

order	who	when
1	stacks (D4 export cap, May-Aug)	every 5 s: the export above 18 kW, up to ~12 kW charger
2	R290 boost (r290_boost.py, .218)	cron 1/min: PCC ≥ 18 kW -> hot water 55 deg C
2	attic air conditioning (hantech.py, .218)	cron 1/min: PCC ≥ 18 kW -> on
3	D5 DO4 pulse -> pv_relay/DO4 (FoxESS shedding)	PCC > 20 kW for 2 min (DO4_GRACE_S), at once > 23 kW (DO4_HARD_W); "1", 3 s later "0"; at most every 5 min; batmonitor/do4_last retained
protection rule (0.51-c), separate from the order above		
!	D5 AC-out overfeed -> DO4 at once	only with the grid offline (vebus /Ac/ActiveIn/Connected != 1): PV feeds > 5 kW into one MultiPlus AC-out for 10 s (factor 1.0 rule, 5000 VA per unit; FoxESS on AC-out1 above 15 kW); with the grid there: bypass via the 50 A transfer relay

forecast scan (ladesperre wrapper, not a decision)
now .. 21:00 in 15 min: PV = clear sky x OWM slot kt x correction,
anchored to the measured Sofar PV of now (fades over 1.5 h)
expected export = PV - HOUSE_EST 1000 W; peak_h, win_end_h (last export > cap)

surplus (chain_mode)
Bat1_eff = Bat1 discharge x 1, Bat1 charge x BAT1_CHARGE_FACTOR 0.5
surplus = PCC + Bat1_eff + own_charge - own_discharge
0.59-c: the real PCC in every season (removed: Z2 winter rule, transition cap, forecast rule, WPPV)

bm_d4 bits	
enabled	BATMONITOR_LADESPERRE and EXPORT_CAP > 0 (18000 W)
season	May..August
fc_cap_ahead	forecast export > EXPORT_CAP before 21:00
surplus_cap_on / hold	surplus - EXPORT_CAP > 200 W / > 20 W
chg_prev	a stack charged last cycle
new_day, pcc_peak	display: PCC > 20 kW seen today

bm_d4 state: peak_today (display)
cap_active = enabled and season and fc_cap_ahead
charge above the cap = surplus_cap_on or (chg_prev and surplus_cap_hold)
block (wait, LADESPERRE 0 W) = cap_active and not charge above the cap
cap_active -> charge target PCC = +EXPORT_CAP (rule text CAP)
[make check: 384 combinations, 48 cap, 24 waiting; fixed point]

block

replaces 0.48-c latch / ratio / bad weather
(only 12 of 4096 combinations blocked, one cloudy
cycle ended the window for the day); the anchored
forecast follows clouds within one cycle

D2 bm_d2 per stack state: prot, force, low, chg_prev
charge_mode: surplus > 200 W on, held while > 20 W
first match: BMS_MISSING > FORCE (force or low) > STALE > LADESPERRE (charge_mode and block) >
CHARGE (SoC < 100, or force_full / top hold window) / FULL > HOLD > PROT > CHARGING > DIS
force_full_from_pv (0.62-c, MQTT S1 / S2): that stack charges first (FF), stays in CHARGE at 100 %,
never discharges (HOLD without surplus); ends at 60 min at 100 %, cell spread ≤ 30 mV or after 240 min
top hold (0.73-c, TOP_HOLD_STACK): window CHARGE at 100 % (TOP) once per day until 240 min / spread end
LOW (0.52-c, state low): on at DCL < DCL_FORCE_A 1 A or lowest cell < CELL_FORCE_V,
off when DCL is back, the cell ≥ CELL_FORCE_RELEASE_V 3.25 V and (0.55-c) SoC ≥ SOC_MIN_RELEASE;
LOW -> FORCE (charge, no discharge); Lenggries env: CELL_FORCE_V 3.05 V / release 3.20 V
[make check, CBMC h_d2_bits]
make check: 64512 combinations per stack (336 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 32256 · FORCE 23040 · STALE 4608 · LADESPERRE 1152 · HOLD 1152 ·
CHARGE 1080 · PROT 432 · CHARGING 360 · DIS 360 · FULL 72

mode per stack

mode matrix BM_MODE_RULE[mode] (0.48-c): what each mode means for the setpoint

mode	action	capped	rule text
BMS_MISSING	0 W	-	NAME
FORCE	FORCE_CHARGE_W 500	**1** (0.48-c)	FORCE(soc %) [LOW]
STALE	0 W	-	NAME
LADESPERRE	0 W	-	LADESPERRE(peak h)
CHARGE	charge amount	1	CHARGE SRC:.. [FF] [CAP] [TOP]
FULL	0 W	-	NAME
PROT	0 W	-	PROT(soc %)
CHARGING	0 W	-	CHARGING(BMS W)
DIS	discharge amount	1	(S4: PROP / IDLE / HOUSE / EXPCAP)
HOLD	0 W	-	HOLD(soc %)

chg_cap per phase (one per stack)
min(CHARGER_A 70 A x U / CHARGE_EFF, CHARGE_MAX_PHASE 4900, BMS CCL)
~4.0-4.2 kW per phase -> ~12 kW for the three Multis
top hold (0.73-c): TOP_HOLD_STACK capped so its highest cell stays at TOP_HOLD_CELL_V 3.55 V
(integral on the cell error, rule tag CELL)

FORCE

CHARGE

FORCE: min(FORCE_CHARGE_W, chg_cap) per phase
from the grid, obeys the BMS CCL (was: 500 W even at CCL 0)
at any time of day: no TOU window (removed 2026-10-11)

charge amount (chain_charge, whole watts since 0.48-c)
want = max(0, own_charge + KP_PM 1010 x (PCC + Bat1_eff - target) / 1000)
target = SOYO_TARGET + CHARGE_RESERVE (0.72-c, Lenggries env 100 W), + EXPORT_CAP while D4 cap_active
at most the sum of chg_cap over the charging phases
[CBMC h_charge]

0 W rows

alloc_levels (0.48-c: one split core for charge and discharge)
charge: one level per stack in CHARGE_PRIORITY order (Stack1, Stack2), the lead stack last
(force_full: its stack first),
each up to min(chg_cap, CHARGER_CAP_PHASE); water filling within a level;
only what the caps cut goes on; a rest up to CHARGE_MAX_PHASE on all charging phases
[CBMC h_charge without bank assumption, h_alloc_discharge]

setpoints per phase (+ = charge) -> Multis
rule text: CHARGE SRC:.. BAL/EQ/FF [CAP] [TOP] [CCL / CELL] | LADESPERRE(h) | FULL | HOLD(soc) | FORCE(soc) [LOW]

clear summer day: how the export cap keeps the PCC below 20 kW			
time	**PV / PCC**	**batmonitor**	**stacks**
morning	export below the cap	LADESPERRE (wait)	0 W, stay free
midday .. afternoon	export above 18 kW	CHARGE CAP: PCC held at 18 kW	take the export above the cap
stacks at their charger limit	export still above 18 kW	R290 boost + air conditioning; DO4 after 2 min > 20 kW	full charge power
after the forecast window	export below the cap again	CHARGE (no cap)	fill up to 100 %
cloudy forecast	no export > cap expected	no cap	charge at once
clouds come up	measured PV low	anchor drops -> cap ends	charge at once